DEEPFAKE X — FORENSIC MEDIA INTELLIGENCE REPORT

CRYPTOGRAPHICALLY AUDITABLE MULTIMODAL AUTHENTICITY EXAMINATION

Investigation ID: DFX-1794449085-384CBB	**Date/Time:** 2026-11-12 02:04:45 UTC
Case Title: Automated Phase 5 Verification Test	**Investigator:** Evaluation Agent
Overall Assessment: No strong evidence of manipulation detected	**Custody Standard:** ISO/IEC 27037 Compliant

1. EXECUTIVE EVIDENTIARY ASSESSMENT

Final Finding: No strong evidence of manipulation detected

Technical Rationale: All 3 evaluated media item(s) predict authentic within standard decision thresholds. No overt generative artifacts or cross-modal stream length discrepancies were discovered.

2. EVIDENCE INVENTORY & CHAIN OF CUSTODY (SHA-256)

Filename	Modality	Size	SHA-256 Digest
real_sample.jpg	Image	0.48 MB	20c4fb5a46856d29e129568d2b4e26281eacf274b6444383e91d822c553cb77a
authentic_video_short.mp4	Video	0.38 MB	1c90590da649b20a7869b989c860c3a782638520d3d1ad31d0f909ddc9655461
real_yt_0000_part_001.flac	Audio	0.40 MB	a986e4cc4a3689533608307f477cdb43979784c58e1711b3db6620b96aa115e5

3. DETECTOR INFERENCE FINDINGS (UNCALIBRATED)

Image Analysis (Swin Transformer Vision Backbone):
• **real_sample.jpg**: Model predicts authentic | Synthetic: 40.09% | Authentic: 59.91% | Device: NVIDIA GeForce RTX 2050

Video Analysis (Uniform Frame Screening):
• **authentic_video_short.mp4**: Flagged 0 / 4 frames synthetic | Mean: 0.97% | Peak: 1.16% at 00:07.50

Audio Analysis (Wav2Vec 2.0 Sequence Classifier):
• **real_yt_0000_part_001.flac**: Model predicts Authentic Human Speech | Synthetic: 0.0% | Authentic: 100.0%

4. CROSS-MODAL EVIDENCE CORRELATION MATRIX

Media Pair	Correlation Type	Agreement	Key Observation
real_sample.jpg ↔ authentic_video_short.mp4	IMAGE_VIDEO_CROSS_INSPECTION	CONCURRING	Classifiers concur (Both predict authentic).
real_sample.jpg ↔ real_yt_0000_part_001.flac	IMAGE_AUDIO_MULTIMODAL_CONCURRENCE	CONCURRING	Modality alignment: Both predict authentic
authentic_video_short.mp4 ↔ real_yt_0000_part_001.flac	VIDEO_AUDIO_STREAM_CORRELATION	CONCURRING	POTENTIAL STREAM DESYNC: Significant duration disparity (5.36s delta) between video and audio track.

5. FORENSIC METHODOLOGY & RELIABILITY LIMITATIONS

ISO/IEC 27037 Evidentiary Standard Notice: This automated analysis report is generated for investigative screening and prioritization. Scores represent raw uncalibrated softmax activations of deep neural feature extractors. 1. **Image Detection:** Standard vision classifiers may exhibit false negatives on novel diffusion generators (e.g. Flux, SDXL).
2. **Video Screening:** Performs frame-level artifact analysis; does not constitute face-swap or lip-sync biometric proof.
3. **Audio Detection:** Wav2Vec2 audio models are sensitive to acoustic channel noise, bandpass filtering (e.g. historical radio), and can fail on state-of-the-art flow-matching voice clones (e.g. ElevenLabs).
4. **Correlation:** Inter-modal correlation does not prove intentional manipulation. Conflicting signals must be verified manually.